allometrik

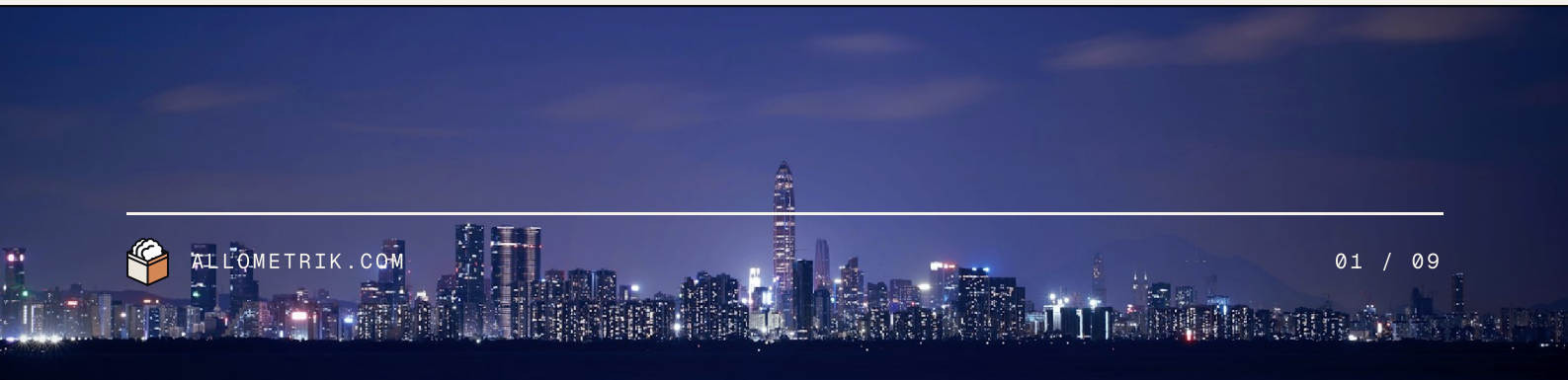
FREE GUIDE · FIELD NOTES FROM SHENZHEN

The AI Toys *Boom*

How talking plush, pendants and desk robots went mainstream, what is inside them, and what it means for builders everywhere.

01	The market in numbers
02	Anatomy of an AI toy
03	Who is building
04	How Shenzhen goes from design to shelf
05	Risks and open questions
06	What this means for builders outside China

Plain English, no hype. Facts checked 3 November 2026; sources on page 8.



Toys became the first mass-market home for *talking AI hardware.*

An AI toy is a plush, pendant or small robot with a microphone, a speaker and a link to a language model. It talks, remembers and reacts. China makes roughly 70% of the world's toys, so the category scaled there first.

¥29 bn China AI toy market in 2025, per MIIT. Forecast: above ¥34.2 bn in 2026	**¥740 m** tracked online sales in 2025, up 427% (Moojing). 1.6 m units
+184% online sales growth in H1 2026: ¥850 m in six months (Moojing)	**¥340** average selling price by Nov 2025, down from over ¥1,000 early that year

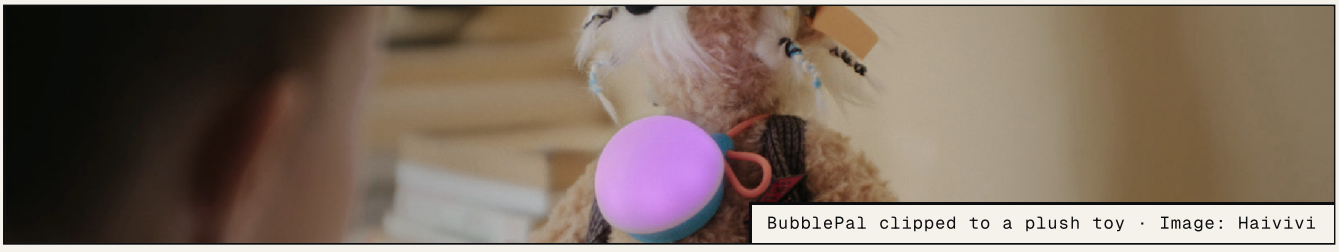
Two measures	MIIT's figure covers the whole market; Moojing tracks e-commerce sales only. Both point the same way: fast growth from a small base.
Where it's made	More than half of China's smart-toy companies are in Guangdong, and 38.53% of those are in Shenzhen (Shenzhen Toy Industry Association).
Margins	Traditional toys often earn 20–30% gross margin. Basic AI toys at ¥300–400 are reported at around 50–65% (Qianzhan).
Mainstream	On JD.com's 2026 "618" festival, AI toy SKUs rose 210% and sales volume 283% year on year.

The price fell below an impulse-buy threshold, and volume followed.

Cheap hardware outside, *the cloud inside.*

Most products follow the same pattern. The toy captures and plays audio; the thinking happens on a server.

LAYER	WHAT IT IS · EXAMPLE FROM A BUBBLEPAL TEARDOWN
Brain chip	A low-cost Wi-Fi/Bluetooth microcontroller. BubblePal uses an Espressif ESP32-S3; FoloToy's AI Passport an ESP32-C3.
Mic + speaker	One electret microphone, a tiny 15×11 mm speaker and a Class-D amplifier. Picking up a child's voice in a noisy room is the hard part.
Battery	A small Li-ion cell (720 mAh in BubblePal), USB-C charging and protection chips.
Shell	Plush, silicone or ABS plastic. BubblePal is a 49.5 g clip-on that turns any stuffed animal into the "body".
Cloud model	Speech-to-text, a large language model and text-to-speech run on servers. FoloToy's founder reported 3–5 s replies after moving voice processing off the chip.
Character + memory	Personas, a voice the child likes, and long-term memory of past conversations.
Companion app	Wi-Fi setup, parental controls, content settings and, often, a subscription.



BubblePal clipped to a plush toy · Image: Haivivi

COST LOGIC

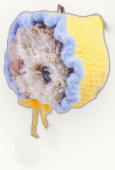
Hardware is cheap: the main chip retails for about ¥7–13. The real cost comes after the sale. Every conversation calls a cloud model, so one-off pricing is under pressure, and many brands add a paid membership after a free period.

Four products that show where the category *is going.*

Figures are as reported by companies or press, not independently audited.



BUBBLEPAL



FUZOZO



LOOI



FOLOTOY

PRODUCT	WHAT HAS BEEN REPORTED
BubblePal Haivivi · Shenzhen	Clip-on pendant that makes any plush toy talk, for children 3+. ¥399. Over 250,000 units and above ¥100 m in sales in under a year (reported June 2025).
Fuzozo Robopoet	AI plush companion for young adults, with eye contact, memory and an evolving personality. ¥399. Over 300,000 units sold, per the developer (Xinhua, Oct 2026).
LOOI TangibleFuture · Shenzhen	Desk robot that uses your phone as its face and brain. About $150. 10,000+ units shipped since November 2024, 90% overseas.
FoloToy FoloToy	AI plush toys and kits, ¥258–699. The Kumma bear sold for $99 abroad. About 20,000 units in 2024.
Others	Ropet (desk pet, nearly 40,000 units), Loona (home robot pet, ~¥3,000, 100,000+), Alpha Group's AI Lazy Goat (100,000+ orders in 2026). Mattel announced work with OpenAI in June 2025; no product yet.

Why a toy idea becomes a product *fastest* in Shenzhen.



Inside SEG Plaza, Huaqiangbei · Photo: Bobbie Johnson, CC BY-SA 2.0

1.45 km²

Huaqiangbei, the electronics market district: 115,000+ businesses, 1 million+ kinds of parts

61%

of electronic products there were AI-related (July 2026), up from 41% in 2025

Parts in an hour	Officials say over 90% of the core parts for a smart device can be sourced within an hour.
Ready-made “cores”	Module makers sell complete AI voice boards for toys. At the 2026 China Toy Expo, one Shenzhen maker said a single AI pet module had sold 100,000 units.
Fast cycles	The fastest path from concept to mass production is about one month for phone-type devices; typical iteration cycles run three to six months.
Global reach	Huaqiangbei electronics go to 190+ countries; for most merchants, foreign trade is over 70% of sales.

The same supply chain also feeds Shenzhen’s robot makers, such as UBTECH, the world’s top humanoid maker by revenue in H1 2026 (Frost & Sullivan).

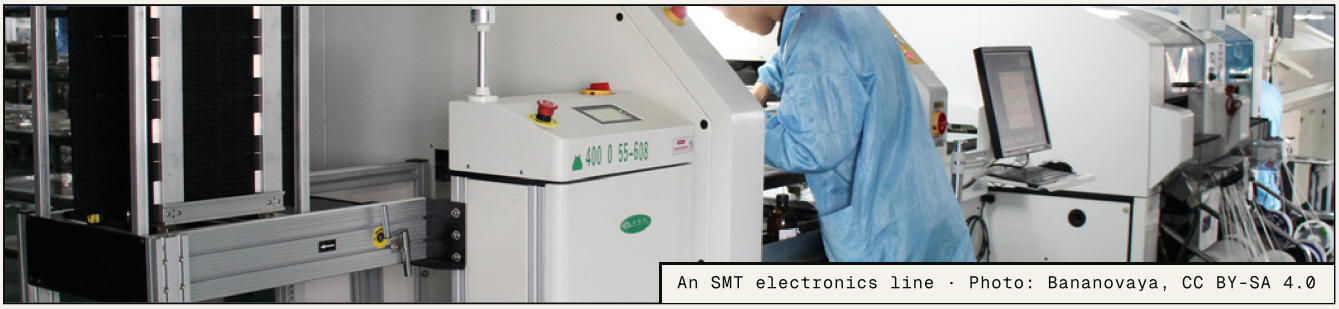
A toy that talks to children *raises the bar.*

The category grew faster than its rules. These are the issues every buyer, retailer and regulator now asks about.

Child safety	In November 2025 US PIRG's "Trouble in Toyland" found FoloToy's Kumma bear gave advice on finding knives, pills and matches and discussed sexual topics. OpenAI suspended the developer; FoloToy paused sales for a safety audit, then resumed.
Privacy	Toys record children's voices and send them to the cloud, sometimes to several AI providers. Push-to-talk, data minimisation and clear parental controls matter.
EU AI Act	Since 2 February 2025 it bans AI that exploits vulnerabilities linked to age to distort behaviour in harmful ways.
EU toy rules	The new Toy Safety Regulation (EU) 2025/2509 entered into force on 1 January 2026 and applies from 1 August 2030, with a mandatory digital product passport for every toy.
China rules	Interim measures on human-like AI interaction services took effect on 15 July 2026; a standard for smart-toy network and data security is being drafted.
Staying power	Many products use similar technology and compete on looks and price. One industry executive cited 30–40% e-commerce return rates.

Open question: will people still talk to these toys after the first month?

Lessons for builders and *companies elsewhere.*



An SMT electronics line · Photo: Bananovaya, CC BY-SA 4.0

01 **The model is not the moat**

Products share the same chips and cloud models. Character, voice, content and trust are what set them apart.

02 **Design for the running cost**

Every conversation costs money. Price a subscription or a usage cap into the plan from day one.

03 **Safety and privacy as features**

With the EU rules above, guardrails, age design and data handling can be what wins the retailer and the parent.

04 **Use the supply chain, own the brand**

Off-the-shelf AI modules and Shenzhen prototyping make hardware fast. Your edge is the IP, the market and the experience.

05 **Beyond children**

Fuzozo and LOOI sell to adults. Companions for adults, older people, learning and the desk are open territory.

Where the numbers *come from.*

Checked on 3 November 2026. Company sales figures are self-reported or press-reported, not audited.

Market size, Fuzozo, LOOI	China Daily via Shenzhen Government Online, Dec 2025, citing MIIT; Cailian Press (cls.cn), 23 Oct 2026, citing MIIT 2025 and 2026 figures, China rules, return rates.
Online sales, prices	Moojing Market Intelligence, “China’s AI Toy Market 2025”; Xinhua, “AI ‘pets’ for China’s lonely…”, 31 Oct 2026 (H1 2026, Fuzozo, Ropet, Loona, Alpha Group, 70% share).
JD 618 2026	China Toy Expo media centre, 18 Aug 2026, citing JD.com.
Shenzhen share, margins	TMTPost, 2025, quoting the Shenzhen Toy Industry Association and Qianzhan. tmtpost.com/7790594
Teardown and hardware	52audio, BubblePal teardown (52audio.com/archives/250572); eefocus, FoloToy AI Passport teardown; FoloToy, “Hello FoloToy” founder notes (folotoy.com).
BubblePal, FoloToy sales	Jiemian, 2025. jiemian.com/article/12961086
Kumma incident	U.S. PIRG Education Fund, “Trouble in Toyland 2025”, 13–14 Nov 2025; NPR, 20 Nov 2025; Futurism.
Mattel and OpenAI	Axios, 15 Dec 2025.
EU rules	EUR-Lex, Regulation (EU) 2025/2509 (OJ 12 Dec 2025); EU AI Act, Regulation (EU) 2024/1689, Art. 5.
Huaqiangbei	Xinhua, 30 Oct 2026; People’s Daily Online via sz.gov.cn, 25 Aug 2026.
Humanoid ranking	Amazing Shenzhen, Oct 2026, citing Frost & Sullivan.

Image credits. Skylines (pages 1, 9): The Agent Camp. Huaqiangbei: Bobbie Johnson via Wikimedia Commons, CC BY-SA 2.0. SMT line: Bananovaya via Wikimedia Commons, CC BY-SA 4.0. Product cutouts: Haivivi, Robopoet, TangibleFuture, FoloToy official websites; editorial identification only.

A NOTE TO FINISH

Want to see it *in person?*

We’re taking a small group (max 10) to Shenzhen in April 2027.

Message me “SHENZHEN” or write to hello@allometrik.com

